

Bi-Molecular Coupling-Energy Challenge: Kernel Ridge Regression on Supermolecule Coulomb Matrices

Challenge in Bi-Molecular Machine Learning — Peter Zaspel, Bergische Universität Wuppertal, HPC group

Task & data

Predict the excitonic coupling energy of a pair of molecular conformers (A, B) from their 3D geometries. The dataset (BiMolData) provides 200 conformers each of monomer A and monomer B — verified to be the same 15-atom molecule ($C_6H_5NO_3$) in identical atom ordering across all conformers — together with coupling energies for the full $200 \times 200 = 40000$ pair grid. `Coord_supermol.xyz` was verified to be exactly the concatenation of the paired A and B geometries at their true relative arrangement, atom-for-atom, with zero numerical difference.

Representation

The bi-molecular difficulty is that the target depends on the shape of A, the shape of B, and their relative arrangement. Rather than combining two separate per-monomer representations, we build a single Coulomb matrix (Rupp et al., *PRL* **108**, 058301, 2012) for the 30-atom "supermolecule" formed by A and B at their true relative geometry — exactly what `Coord_supermol.xyz` encodes. Its off-diagonal entries are $M_{ij} = Z_i Z_j / |R_i - R_j|$, diagonal $M_{ii} = 0.5 Z_i^2$. This single 30×30 matrix contains the A-A block, the B-B block, and the A-B (inter-monomer) block that directly drives a Coulombic/excitonic coupling — no separate combination scheme is required. The upper triangle (465 entries) is used as the feature vector. We skip the usual row/column sort-by-norm (permutation-invariance) step: every conformer shares an identical atom order, so there is no permutation ambiguity to remove, and skipping it keeps each feature tied to the same physical atom pair across all samples.

Model & hyperparameter selection

Kernel Ridge Regression with a Laplacian kernel, $k(x, x') = \exp(-||x - x'||_1 / \sigma)$, the classical Coulomb-matrix pairing from Rupp et al. A fixed 10% test set (4000 pairs) was held out up front and never used for hyperparameter selection or training. Hyperparameters ($\gamma = 1/\sigma$ and ridge regularizer λ) were chosen by 5-fold cross-validated grid search (log-spaced grids, $\gamma \in [10^{-4}, 10^{-1}]$, $\lambda \in [10^{-6}, 10^{-1}]$) on a 2000-sample subset of the training pool, selecting $\gamma = 10^{-4}$, $\lambda = 10^{-6}$ (CV MAE 2.11), then held fixed across the learning curve to bound total compute (KRR fitting is $O(N^3)$).

Validation / learning curve protocol

For each training size N in a log-spaced sequence (50 to 12800), we drew random training subsets from the 36000-pair pool (3 repeats for $N \leq 3200$, 2 for $N = 6400$, 1 for $N = 12800$, reduced at large N purely to bound wall-clock time), fit KRR with the fixed hyperparameters, and measured MAE against the fixed held-out test set. Mean MAE (min/max across repeats) vs. N is plotted on log-log axes at right.

Results & discussion

Test MAE falls from 13.1 ($N=50$) to 0.175 ($N=12800$) — a clean, near power-law decay, $MAE \approx N^{-0.79}$ (log-log least-squares fit, $R^2=0.93$; see Fig. 1 and table). For reference, the target's standard deviation is 38.4, so even the smallest model ($N=50$) already explains most of the variance, and by $N=12800$ the error is $\approx 0.5\%$ of the target spread. The challenge notes kernel-based models can reach MAE in 0.01–0.1; the observed trend is consistent with reaching that range at somewhat larger N (the curve shows no sign of flattening within the range explored).

The learning curve follows the expected power-law decay of MAE with training-set size for kernel-based molecular-property regression. The main limitation is KRR's $O(N^3)$ training cost, which caps the largest training size explored here well below the full 36000-pair pool; a Nyström approximation or inducing-point Gaussian Process would be the natural next step to push further, and re-tuning λ per point (rather than once, globally) would likely lower the curve slightly at large N .

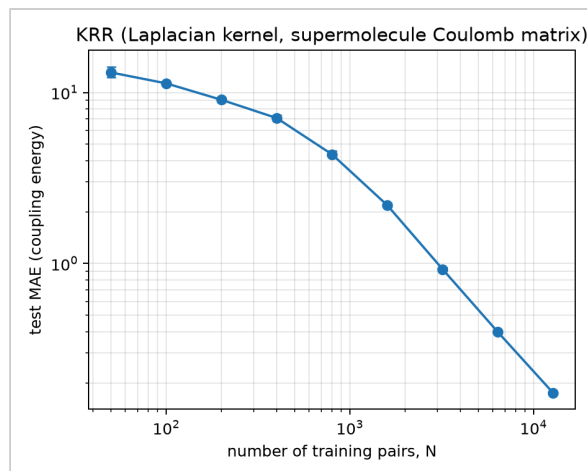


Fig. 1 — Test MAE vs. number of training pairs (log-log), KRR with Laplacian kernel on the supermolecule Coulomb-matrix representation. Error bars show min/max across repeats.

N (train pairs)	MAE
50	13.094
100	11.310
200	9.062
400	7.086
800	4.348
1600	2.190
3200	0.926
6400	0.397
12800	0.175